

NEON SURVIVAL ARENA

Cyber Survivor — Top-Down Neon Survival Shooter

Game Design Document & Technical Walkthrough • GitHub Portfolio Project

THE PITCH

You open the website. A glowing neon world appears. You tap **Play**. Your character is a glowing blue orb. Enemies begin walking toward you. You move, you aim, the game auto-fires. Kill, collect XP, level up, pick an upgrade, get stronger — while the arena keeps getting harder. Every 3 minutes a boss appears. There is no ending. Just "I survived 18 minutes... I can beat 20."

FIELD	DETAIL
Genre	Top-down survival shooter (roguelite)
Platform	Single web codebase — desktop + mobile
Engine	HTML5 Canvas + vanilla JavaScript (no dependencies)
Session	Endless runs, ~5–25 min each
Persistence	Browser localStorage (scores, coins, unlocks)

1. Controls — One Codebase, Two Schemes

The game detects touch capability at load and swaps the input layer. Core game logic is identical; only the control adapter changes.

Mobile (twin-stick)

- **Left thumb** — virtual joystick for movement.
- **Right thumb** — aim joystick for direction.
- **Auto-shoot** — the player only worries about moving and aiming.

Desktop

- **WASD** — movement.
- **Mouse** — aim direction.
- **Left click** (or auto-fire toggle) — shoot.

2. The Core Gameplay Loop

This loop is the engine of addiction — a constant stream of small decisions and escalating tension.



Every upgrade changes how the game *feels*. Because the player picks 1 of 3 random upgrades each level, no two runs play the same.

3. Upgrade System

On every level-up, present 3 random upgrade cards. Some are new weapons, some modify existing ones, some are passive stats. Stacking creates emergent builds.

UPGRADE	EFFECT	TYPE
Fire Bullets	Burn damage-over-time on hit	Weapon mod
Ice Bullets	Slow / freeze enemies	Weapon mod
Lightning Chain	Arcs to nearby enemies	Weapon mod
Explosive Rounds	Area damage on impact	Weapon mod
Ricochet	Bullets bounce to new targets	Weapon mod
Triple Shot	Fire 3 bullets in a spread	Weapon
Drone Companion	Orbiting drone auto-fires	Companion
Health Regen	Slowly recover HP over time	Passive
Faster Movement	+ move speed	Passive
Double Fire Rate	Shoot twice as fast	Passive
Bigger Explosions	+ blast radius	Passive

Design note: keep the 3 cards weighted — early levels favor foundational picks, later levels unlock exotic combos (e.g. Explosive + Ricochet = chaos).

4. Enemy Types

Not just zombies walking forward — the mix forces the player to constantly adapt positioning and targeting priority.

ENEMY	BEHAVIOR
Tiny Fast Bugs	Swarm quickly, low HP, overwhelm by numbers
Slow Tanks	High HP, slow, soak damage and push forward
Snipers	Stay at range, fire aimed shots
Flying Drones	Ignore obstacles, erratic approach
Exploding Robots	Rush the player and detonate on contact
Teleporting Ninjas	Blink toward player, unpredictable
Shield Enemies	Frontal shield — must be flanked

5. Bosses

A boss spawns every **3 minutes**, announced with a deep warning sound and a screen pulse. Each has distinct attack patterns and a health bar.

BOSS	PATTERN
Giant Robot	Rockets + ground slam shockwaves
Cyber Worm	Burrows and erupts along the arena
Alien Queen	Spawns minion swarms, radial projectile bursts

6. Difficulty Scaling

- Enemy spawn rate rises with elapsed time.
- Enemy HP and speed scale on a curve, not a spike.
- New enemy types are introduced at time thresholds.
- Bosses interrupt the rhythm every 3 minutes as a skill check.

7. Juice — Graphics & Sound

The "feel" is where retention lives. Tiny dopamine moments compound.

Visuals

- Dark background; neon blue, purple, pink; orange explosions.
- Glow effects, particle bursts, and screen shake on impacts.
- Floating damage numbers and XP gem pickups.

Audio

- Hit → little pop. Level up → satisfying chime.
- Boss appears → deep warning tone. Explosion → bass hit.
- Generate procedurally with the Web Audio API (no asset files needed).

8. Meta Progression

Coins earned per run persist in localStorage and unlock content between runs — giving long-term reasons to return.

SYSTEM	UNLOCKS / PURPOSE
New Characters	Different starting stats / weapon
New Weapons	Alternate base fire behavior
Weapon Skins	Cosmetic neon variants
Difficulty Modes	Harder scaling for veterans
Daily Challenge	Fixed seed — compete on one shared run
Achievements	Milestone goals (survive X, kill Y)
Leaderboard	Local high scores (global if a backend is added)

9. Endless Mode & Hook

No ending. The run-time survived is the score. The player always thinks "one more run, I can beat my time." That single mindset is the whole retention model.

10. Suggested Extra Features (added value)

- **Weapon evolutions** — max a weapon + a matching passive to fuse a super-weapon.
- **Combo / kill-streak multiplier** that decays if you stop killing.
- **Pause-menu build summary** showing all active upgrades.
- **Screen-clear "nova" pickup** as a rare panic button.
- **Colorblind-safe palette toggle** and screen-shake slider for accessibility.
- **Seeded runs** so a shared link reproduces the same daily challenge.
- **Run recap screen** — time, kills, level, damage, coins earned.

11. Technical Architecture

Suggested structure for a clean, reviewable GitHub repo:

```
index.html – canvas + UI overlay
/src
main.js – game loop (requestAnimationFrame)
input.js – desktop + mobile adapters
entities/ – player, enemy, bullet, boss, gem
systems/ – spawn, collision, upgrade, particles
audio.js – Web Audio sound synthesis
storage.js – localStorage save/load
ui.js – menus, HUD, upgrade cards
/assets – (optional) fonts, icons
```

12. Skills This Project Demonstrates

SKILL	WHERE IT SHOWS
Canvas rendering	Drawing entities, glow, particles each frame
Game loop	Fixed-timestep update + render via rAF
Collision detection	Bullet→enemy, enemy→player, gem pickup
Particle effects	Explosions, trails, screen shake
Pathfinding / steering	Enemies seek the player, avoid stacking
State management	Upgrade system, run state, menus
Web Audio API	Procedural sound synthesis
Persistence	localStorage for scores & unlocks
Responsive input	Unified desktop + touch controls

13. Build Roadmap

PHASE	GOAL
1 — Core	Player moves, auto-shoots, enemies spawn & die
2 — Loop	XP gems, level-up, 3-card upgrade picker
3 — Content	Enemy variety, bosses, difficulty scaling
4 — Juice	Particles, screen shake, Web Audio sounds
5 — Meta	Coins, unlocks, leaderboard, achievements
6 — Polish	Mobile controls, accessibility, README + deploy



